



the epicentre of debates on whether benchmark scores can actually relate to real-world performance. Snapdragon processors are known to be better at multimedia related SIMD processes even though these score

lower at benchmark tests. Snapdragon's hardware acceleration of Adobe Flash and WebGL content delivers a smooth Web experience without the freezing or jerking seen if you have tasks running in the background.

This SoC also utilises a Harvard Superscalar architecture. It comes with embedded processor (supporting GSM, GPRS, EDGE, UMTS/WCDMA, HSDPA 7.2Mbps, HSUPA 5.76Mbps, HSPA+ 28Mbps/11Mbps, MBMS baseband), embedded seventh-generation gpsOne GPS module and gpsOneXTRA Assistance.

Apple A5 800MHz

Designed by Apple and manufactured by Samsung using a 45nm process, this is the latest ARM processor



from Apple featured in its iPhone 4S and iPad 2 devices. A5 is a chip based upon the dual-core ARM Cortex-A9 MPCore CPU with NEON SIMD accelerator and a dual-core

PowerVR SGX543MP2 GPU clocked at 200 MHz. Apple lists the A5 to be clocked at 1 GHz on the iPad 2's technical specifications page, though it can dynamically adjust its frequency to economise on battery life. The SoC also contains 512 MB of DDR2 memory clocked at 533 MHz.

This chip includes an image signal processor unit that does advanced image post-processing such as face detection, white balance and automatic image stabilisation. Apple remains the only company to have a PowerVR SGX543 graphics solution on its SoC. This solution makes the A5 chip a lot more expensive than the rest.

TI OMAP4460 1.2GHz

OMAP 4460 features dual-core ARM Cortex-A9 MPCore built using a 45nm process

just like the Apple A5. One difference is that this OMAP is clocked much higher (1.2GHz) than the A5 (800MHz). TI's datasheet specifies that OMAP 4460 can handle up to 1.5 GHz but it's clocked at 1.2



GHz to keep battery consumption within limits. The GPU featured here is the PowerVR SGX540, which helps in 2D and 3D graphics acceleration.

This SoC also features TI's Smartflex technology that reduces power consumption by dynamically controlling the voltage, frequency and power based on the device activity, modes of operation and temperature. It can handle 1080p HD video as well as 1080p stereoscopic 3D video.

Upcoming silicon

The upcoming chips show a trend wherein performance alone ceases to be the driving factor. Mindsets have changed and battery life is just as important as performance. The new chips deliver by featuring low-power cores intended to take on the processing strain in stand-by mode.

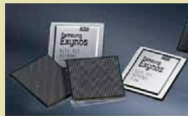
Samsung Exynos 5250

Samsung's Exynos 5250 packs two ARM Cortex-A15 processors clocked at 2 GHz. The dual-core chip is claimed to offer about twice the CPU performance of existing products that are equipped with a pair of ARM 1.5GHz Cortex-A9 processors. It is built using a 32nm low-power high-K metal gate process.

TI OMAP5430

The flagship of TI's new OMAP5 series, this 28nm SoC integrates two ARM Cortex-A15 MPCores that are clocked at

2 GHz each, two Cortex-M4 cores that are used as accelerators and CPUs that power the device when in low-power/standby mode. According to TI, the A15 architecture is about 50 per cent faster than the preceding A9 cores and the entire chip is about three times faster than the previous generation (OMAP4, which isn't available yet).



The graphics engine is based on the PowerVR SGX544-MPx core as well as TI's own 2D graphics engine. Users can run 1080p video at 60 frames per second and convert 2D 1080p video into 3D in 1080p in real time. TI claims that it has enough horsepower to support a 2D digital camera with up to 24-megapixels resolution or 12-megapixels 3D